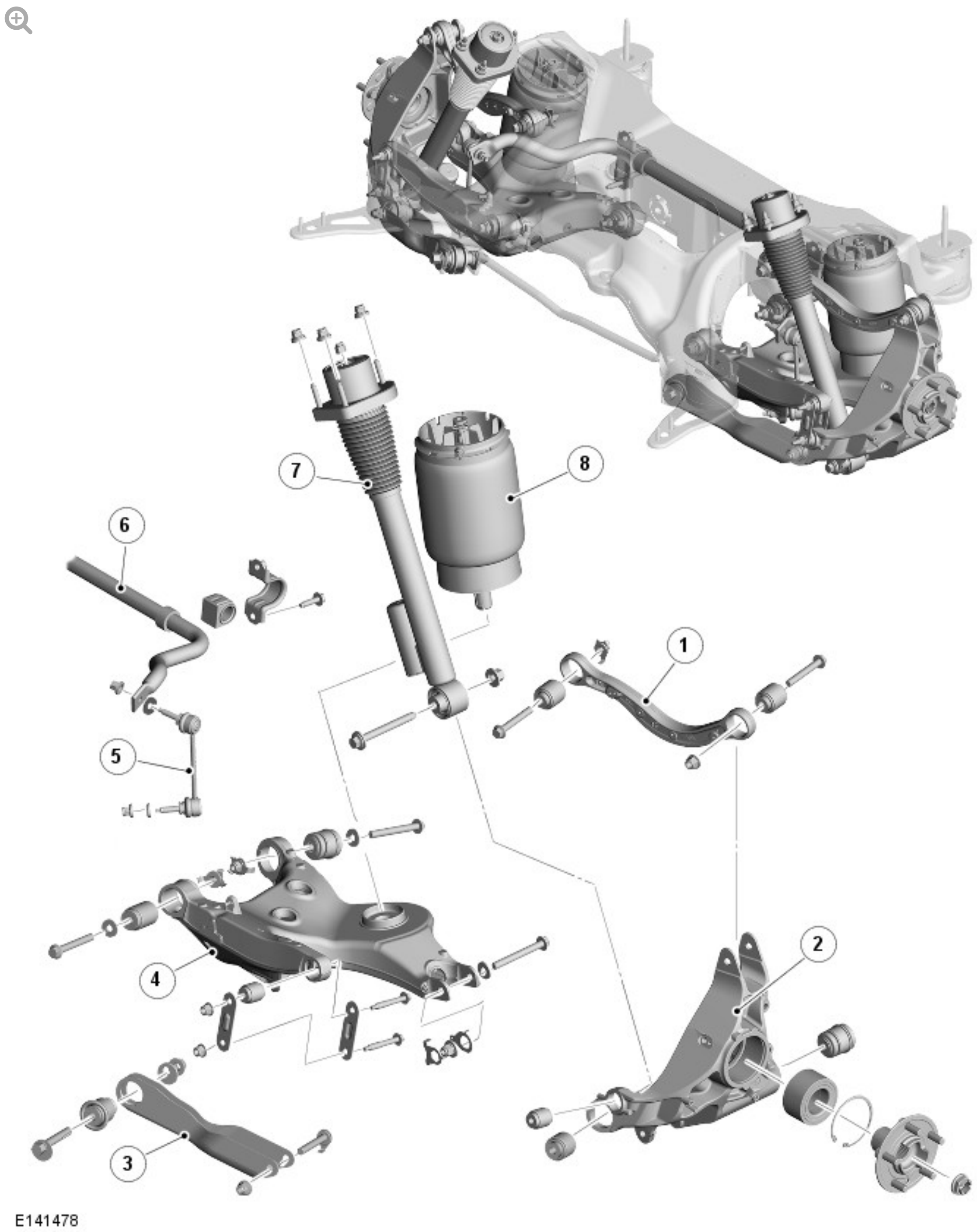
2016.0 RANGE ROVER (LG), 204-02

REAR SUSPENSION

DESCRIPTION AND OPERATION

COMPONENT LOCATION

Rear Suspension Component Layout



ITEM	DESCRIPTION
1	Upper arm
2	Wheel knuckle
3	Tie arm
4	Lower arm
5	Stabilizer link
6	Stabilizer bar

7	Damper
8	Air spring

OVERVIEW

The independent rear suspension offers a reduction in unsprung weight.

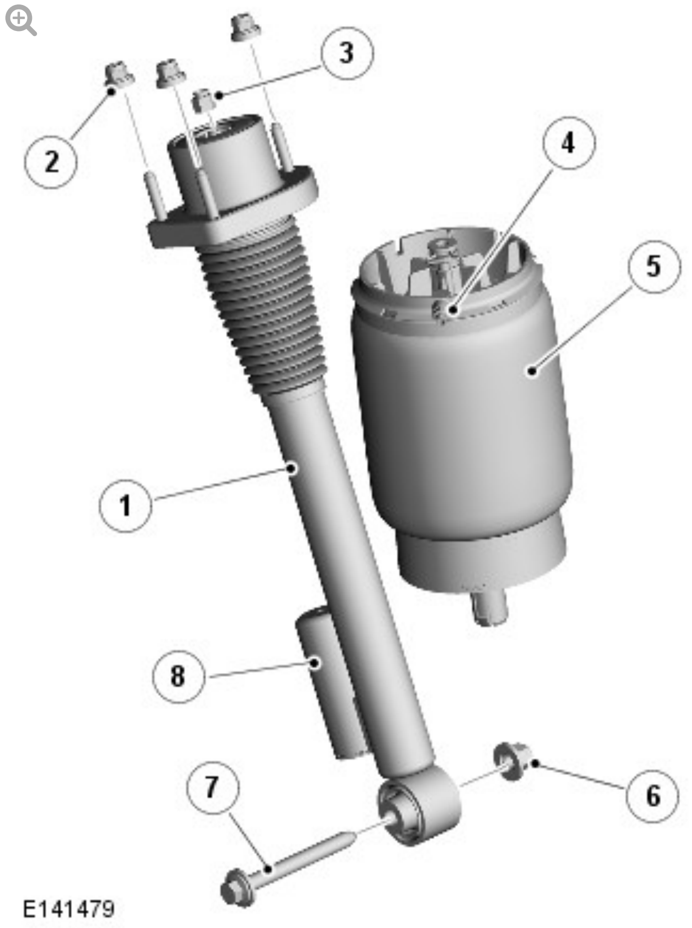
Combined with an aluminum subframe the suspension offers: maximized dynamic performance for ride, handling and steering refinement and minimized vibration.

Depending on vehicle specification, either:

- a conventional stabilizer bar is fitted, or
- a dynamic response stabilizer bar.
  For additional information, refer to: Ride and Handling Optimization (204-06, Description and Operation).

DESCRIPTION

AIR SPRING AND DAMPER



ITEM	DESCRIPTION
1	Damper
2	Nut (damper to vehicle body)
3	Nut (damper rod)
4	Air connector
5	Air spring
6	Nut (damper to wheel knuckle)
7	Bolt (damper to wheel knuckle)
8	Damper - external oil reservoir

DAMPER

The damper assembly is a mono tube design, the lower end of the damper is fitted with a bush and is attached to the wheel knuckle. The damper top mounting is located in a turret in the body and secured to the body with three nuts

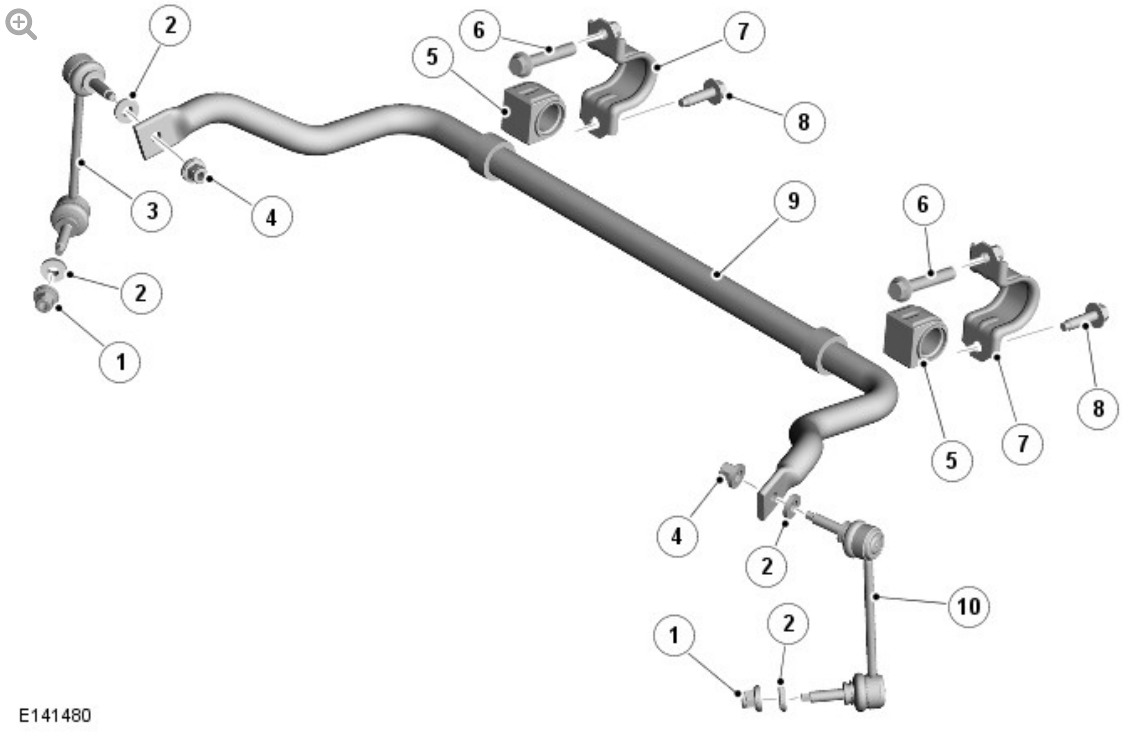
secured to the body with three nuts The damper functions by restricting the flow of hydraulic fluid through internal galleries within the damper. The damper rod moves axially within the damper, its movement limited by the flow of fluid through the galleries, providing damping of undulations in the terrain. The damper rod is sealed at its exit point from the damper body to maintain the fluid within the unit and to prevent the ingress of dirt and moisture. The seal also incorporates a wiper to keep the rod clean.

AIR SPRING

Each air spring comprises a top cap assembly, an air bag and a base piston. The air bag is attached to the top cap and the piston with crimp rings. The air bag is made from a flexible rubber material which allows the sleeve to roll up and down the air spring piston as the vehicle changes height. The top cap assembly comprises the plastic top cap with a spigot which protrudes through a hole in the subframe. This spigot includes a timing feature to ensure the air spring is in the correct orientation and is attached to the subframe with a spring clip. On the side of the top cap is a connector which allows for the attachment of the air pipe from the valve block. The piston is made from plastic and is shaped to allow the air bag to roll over its outer diameter. The base of the piston has a spigot with an integral clip to locate into the lower wishbone.

The air springs are located inboard of the dampers and are retained between the subframe and the lower wishbone.

STABILIZER BAR AND LINKS



ITEM	DESCRIPTION
1	Nut (stabilizer link to lower arm)
2	Washer
3	Stabilizer link (right)
4	Nut (stabilizer link to stabilizer bar)
5	Bush (stabilizer bar to subframe)
6	Bolt (subframe to stabilizer bar bracket)
7	Bracket (stabilizer bar to subframe)
8	Bolt (stabilizer bar bracket to subframe)
9	Stabilizer bar
10	Stabilizer link (left)

Vehicles without the Dynamic Response stabilizer bar system use a conventional stabilizer bar. The Dynamic Response system is detailed in a separate section

For additional information, refer to: Ride and Handling Optimization (204-06, Description and Operation).

The stabilizer bar is fabricated from induction hardened, solid spring steel bar. The stabilizer bar operates, via a pair of links, from their attachment to the lower arm.

The stabilizer bar is mounted on the subframe’s rear cross member and is attached with two, Teflon lined bushes. Brackets, which are pressed onto the bushes, are attached to the cross member with bolts. The stabilizer bar has crimped, 'anti-shuffle' collars pressed in position on the inside edges of the bushes. The collars prevent sideways movement of the stabilizer bar.

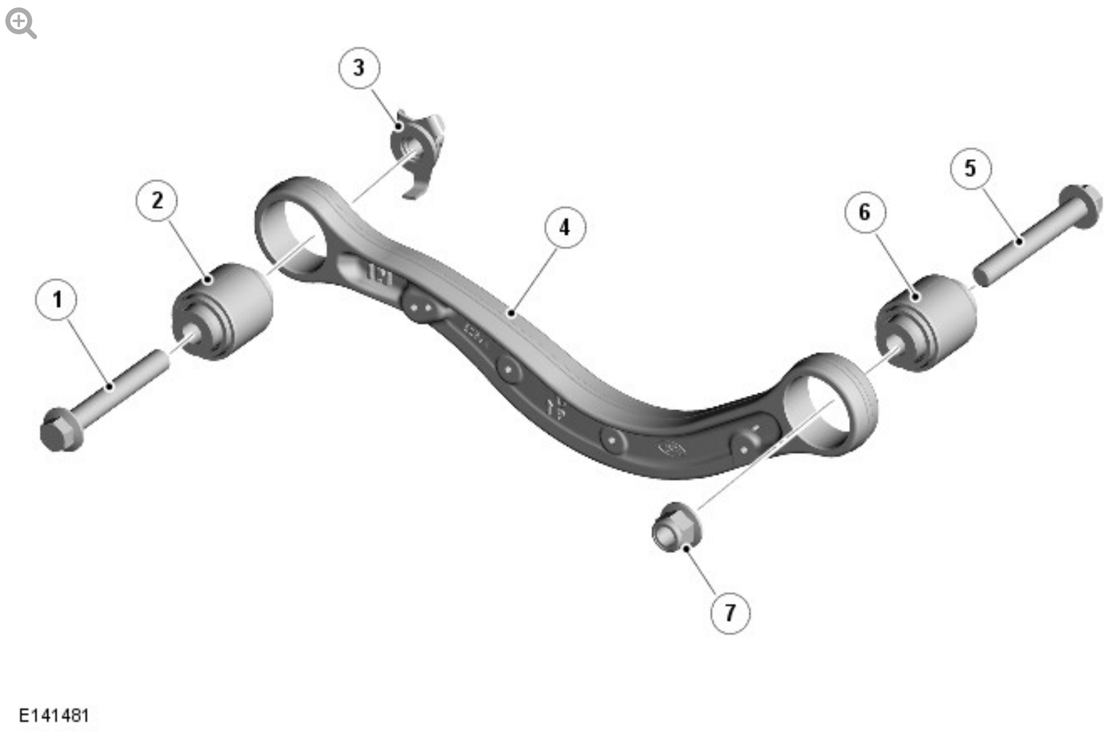
The links have a ball joint at each end, are fitted with hardened steel washers at both ends. The bottom ball joint is attached to the link at 90 degrees to the link axis. The ball joint is located in a hole in the lower arm and secured with the hardened steel washer on one side of the interface and a self-locking nut on the other. The top ball joint is attached to the link at 90 degrees to the link axis. The ball joint is located in a hole in the end of the stabilizer bar and secured with the hardened steel washer on one side of the interface and the self-locking nut on other.

It is important that the hardened steel washer is in the correct position between the stabilizer bar and the link ball joint and the damper yoke and the link ball joint and the correct, hardened washer is fitted.

CAUTION:

Failure to fit the washer or using an incorrect washer will result in relaxation of the torque on the self-locking nut and damage will be caused to the stabilizer bar, link and lower arm.

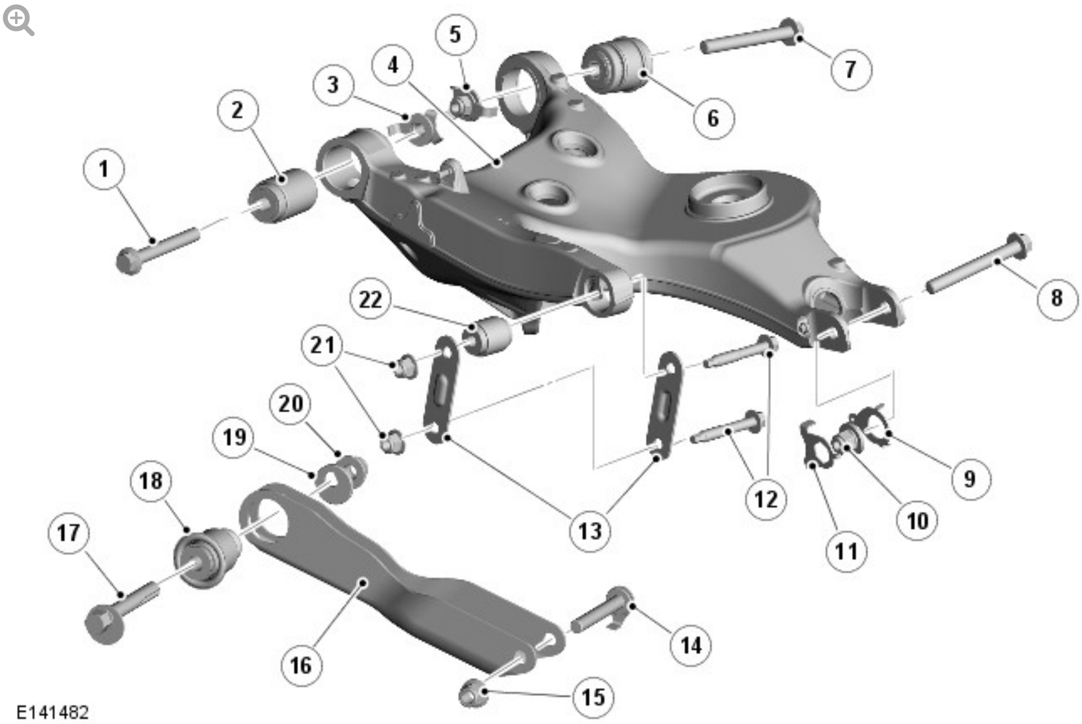
UPPER ARM



ITEM	DESCRIPTION
1	Bolt (upper arm to subframe)
2	Bush (upper arm to subframe)
3	Nut (upper arm to subframe)
4	Upper arm
5	Bolt (upper arm to wheel knuckle)
6	Bush (upper arm to wheel knuckle)
7	Nut (upper arm to wheel knuckle)

The upper arm is made from forged steel with both ends of the arm having a bush housing machined in the arm; a bush is pressed into each of the housings. The inner end locates in brackets on the upper section of the subframe. The outer end of the bar locates in lugs at the top of the wheel knuckle.

LOWER ARM AND TIE ARM



ITEM	DESCRIPTION
1	Bolt (lower arm to subframe)
2	Bush (lower arm to subframe)
3	Nut and washer (lower arm to subframe)
4	Lower arm
5	Nut and washer (lower arm to subframe)
6	Bush (lower arm to subframe)
7	Bolt (lower arm to subframe)
8	Bolt – (lower arm to wheel knuckle)
9	Washer – (lower arm to wheel knuckle)
10	Nut – (lower arm to wheel knuckle)
11	Washer – (lower arm to wheel knuckle)
12	Bolt - link (lower arm to wheel knuckle)
13	Integral link (lower arm to wheel knuckle)
14	Bolt (tie arm to wheel knuckle)
15	Nut (tie arm to wheel knuckle)

16	Tie arm
17	Bolt – toe adjustment (tie arm to subframe)
18	Bush (arm to subframe)
19	Washer – toe adjustment (tie arm to subframe)
20	Nut – toe adjustment (tie arm to subframe)
21	Nut - link (lower arm to wheel knuckle)
22	Bush - link (lower arm to wheel knuckle)

LOWER ARM

The lower arm is cast aluminum with two bushes pressed into the inner section of the arm which provides for the attachment to the subframe. The bushes are located between brackets on the subframe and secured with bolts and nuts. The lower arm has a platform which provides for the attachment of the air spring and a location hole for the attachment of the stabilizer link.

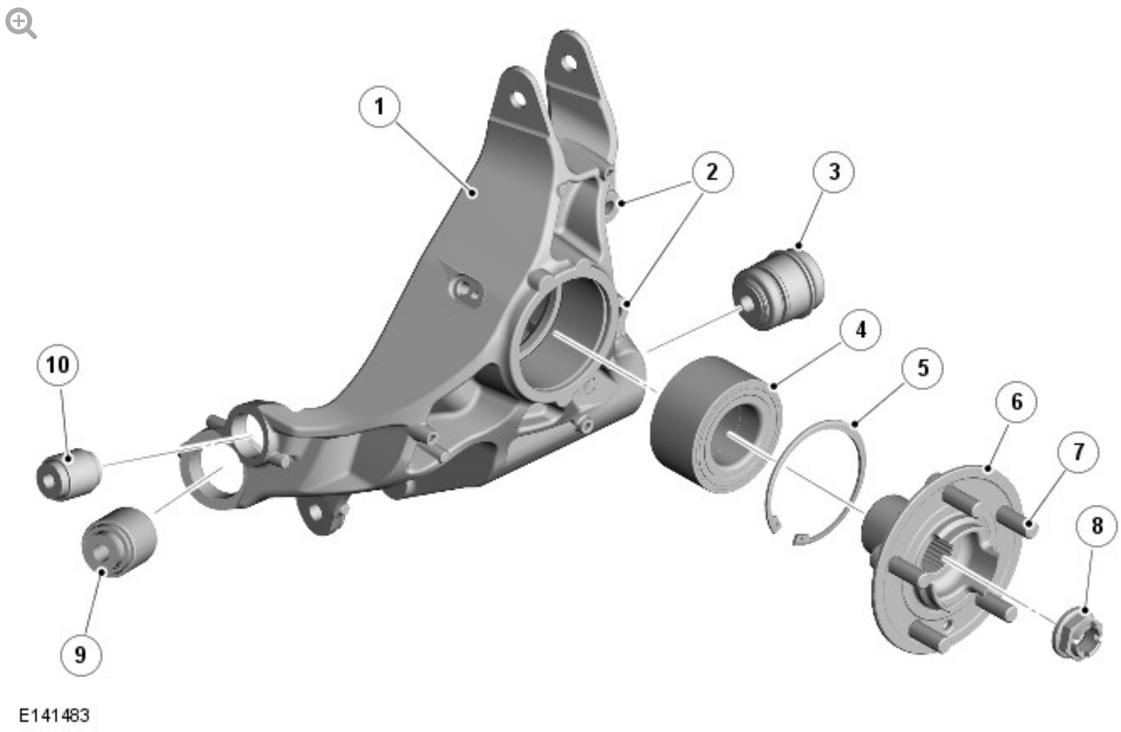
The outer section of the arm is attached to the forward end of the wheel knuckle by links which provides castor control. Lugs at the rearward end of the lower arm attach to the wheel knuckle's rear bush.

TIE ARM

The tie arm is pressed steel with both ends having a bush pressed into an integral housing.

The inner bush locates between brackets on the subframe and is secured with a special eccentric bolt, washer and nut. This allows for the adjustment of the rear wheel toe. Rotation of the bolt moves the eccentric head within a recessed slot in the bracket, allowing the toe to be adjusted within the set limits. The outer bush attaches to the wheel knuckle.

WHEEL KNUCKLE, WHEEL HUB AND BEARING ASSEMBLY



ITEM	DESCRIPTION
1	Wheel knuckle
2	Brake caliper attachment location
3	Bush (lower arm to wheel knuckle)
4	Bearing (wheel hub)
5	Circlip (wheel hub bearing)
6	Wheel hub
7	Wheel stud
8	Nut (wheel hub)
9	Bush (tie arm to wheel knuckle
10	Bush - link (wheel knuckle to lower arm)

The hub assembly comprises a wheel hub, drive flange and bearing. A seal and bearing are fitted in the wheel hub and are secured with a circlip. The drive flange has wheel studs attached to it and locates on the splined drive shaft and is secured with a stake nut.